

Task 1 :

Multimeter Real-Life Troubleshooting Scenarios

Objective: Identify three practical scenarios where a digital multimeter is used in computer and smartphone hardware diagnostics.

1. Testing Power Supply Unit (PSU) Voltage Outputs:

- *How it helps:* Measures DC voltage outputs (12V, 5V, 3.3V) on ATX connectors to verify if the PSU is providing stable power to motherboard components.

2. Checking Smartphone Battery Health & Terminal Voltage:

- *How it helps:* Measures battery terminal voltage to confirm whether a smartphone battery holds sufficient operational charge or is completely degraded/dead.

3. Trace and Continuity Testing on Motherboard Circuits:

- *How it helps:* Uses continuity mode (beep test) to detect broken circuit traces or short circuits across micro-components on the motherboard.

Task 2:

POST Card Diagnostic Summary

- **Objective:**

Summarize the operation and problem-identification capability of a PCI/PCIe POST diagnostic card on a dead/no-display motherboard setup.

- **Summary:**

A POST (Power-On Self-Test) diagnostic card plugs into a motherboard's PCI/PCIe slot and displays two-digit hexadecimal codes (e.g., C1, 55, FF) on its LED display during startup. As BIOS tests system components sequentially, the card updates these codes in real time to show progress. In the test video demo, the POST card halted at error code **55 / C1**, immediately identifying a Memory Initialization Failure (RAM Detection Issue) before any output appeared on the monitor screen.

Task 3 :

Software Diagnostic Tool Execution

Objective: Execute a software-based diagnostic utility (HWMonitor / CrystalDiskInfo) to monitor real-time system temperatures, component voltages, and hardware health status.

- **Diagnostic Software Used: HWMonitor** (by CPUID) / **CrystalDiskInfo**

- **System Metrics Recorded:**
 - **CPU Package Temperature:** ~42°C – 48°C
(Normal operational threshold)
 - **Storage Health Status:** GOOD (100% Health, NVMe/SATA SSD)
 - **System Voltages:** Stable 12V, 5V, and 3.3V rail readings

HWMonitor				
File View Tools Help				
Sensor	Value	Min	Max	
SHUBHAM-7				
HP 8723				
Temperatures				
\TZ.CHGZ	43.0 °C	43.0 °C	43.0 °C	
\TZ.BATZ	34.0 °C	34.0 °C	34.0 °C	
\TZ.LOCZ	48.0 °C	47.0 °C	48.0 °C	
\TZ.EXTZ	47.0 °C	46.0 °C	47.0 °C	
\TZ.GPXZ	25.0 °C	25.0 °C	25.0 °C	
\TZ.CPUZ	56.0 °C	55.0 °C	57.0 °C	
Utilization				
Physical Memory Load	52.0 %	50.0 %	55.0 %	
Available Physical Memory	7.6 GB	7.1 GB	7.8 GB	
Virtual Memory Load	53.8 %	52.2 %	55.0 %	
Available Virtual Memory	7.7 GB	7.6 GB	8.0 GB	
Intel Core i5 10310U				
Voltages				
VID (Max)	0.747 V	0.737 V	0.853 V	
Temperatures				
Cores (Max)	55.0 °C	54.0 °C	61.0 °C	
Package	57.0 °C	54.0 °C	61.0 °C	
Powers				
Package	5.17 W	2.96 W	13.04 W	
IA Cores	3.98 W	1.87 W	11.17 W	
GT	0.01 W	0.00 W	1.18 W	
DRAM	0.65 W	0.44 W	2.07 W	
Power Max (PL1)	44.00 W	44.00 W	44.00 W	
Short Power Max (PL2)	51.00 W	51.00 W	51.00 W	
Max Peak Power (PL4)	17.00 W	17.00 W	17.00 W	
Currents				
IccMax	85.00 A	85.00 A	85.00 A	
Clocks				
Cores (Max)	1397.5 MHz	904.5 MHz	2398.8 MHz	
CPU BCLK	99.8 MHz	84.4 MHz	117.6 MHz	
LLC/Ring	1197.9 MHz	995.4 MHz	2101.2 MHz	
Memory	1332.9 MHz	623.3 MHz	1333.4 MHz	
Utilization				
Processor	35.2 %	10.0 %	77.7 %	
SSD 512GB (AA20230921512G423235)				
Temperatures				
Assembly	40.0 °C	39.0 °C	43.0 °C	

Ready

Logging OFF (F5 to start)

NUM

Task 4 :

Comparison of Hardware & Software Diagnostic Tools

Objective: Compare Multimeter, POST Card, and Software Diagnostics in terms of diagnostic scope and example problems detected.

Tool	What it checks	Example Problem Detected
Multimeter	Electrical parameters (DC/AC Voltage, Current, Resistance, and Circuit Continuity across hardware components).	Identifying a faulty Power Supply Unit (PSU) with low/zero output voltage, or finding a broken trace/short circuit on a motherboard.
POST Card	Diagnostic status codes (Hex codes)	Pinpointing a motherboard boot failure or

Tool	What it checks	Example Problem Detected
	output by the BIOS during the Power-On Self-Test (POST) before the OS boots.	"No Display" issue caused by faulty RAM initialization (e.g., Code 55 / C1).
Software Diagnostics (<i>HWMonitor / CrystalDiskInfo</i>)	Real-time system metrics, component health (temperatures, fan speeds, SSD S.M.A.R.T. status, voltages) inside the OS.	Detecting CPU thermal throttling caused by high temperatures (>90°C) or detecting failing sectors/bad health on an SSD/HDD.